# Scores and Calibration Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

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## Scores and Calibration Intuition

> Prepare Scores and Calibration Intuition: check the target, prerequisites, resources and adaptable teaching schedule.

### Teach the learning cycle

Primary target: Interpret **calibration** using groups of forecasts

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

### Retrieval answer guidance

### Train a Classifier

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Connect **training**-only scaling and neighbour **prediction** in one pipeline

Reference example, not the only acceptable wording: fit learns **training** minima and ranges and stores scaled **training** rows with IDs and labels. predict takes a raw feature pair and the fitted state.

Evaluation compares a **prediction** with its reference label afterward. Validation references belong only in evaluation, not in **fitting** or the neighbour pool.

### Decision Trees

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

**Training** labels determined both leaf majorities and the error comparison that selected near rather than tall.

### Leakage

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test **score** uses that set for development, so it is no longer an untouched final evaluation.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

Twenty cases receive **score** .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another **score** group.

### Concept to explain

A model **score** can rank **prediction**s without matching the real frequency of correct outcomes.

| Term | Meaning |
|---|---|
| score | model value used for ranking or confidence |
| calibration | scores align with observed frequencies |
| bin | group of predictions with similar scores |
| reliability | agreement between score and outcome rate |

### Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

### Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

### Preparation for the connected explanation

Teach two 45-minute sessions. This week focuses on **calibration** as a group property. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

### Readiness answer

No. Its interpretation must be established; a threshold can act on a **score** without making it a calibrated probability.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

### End goal for the pictorial case

By the end, I can explain **score**s and **calibration** intuition using a traced example and a stated limitation.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

# Explain the mechanism

> Model Scores and Calibration Intuition using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Trace the information, fitted behavior, and evaluation proof separately.

### Question to explain

QUESTION Bin **prediction score**s and compare average **score** with observed correctness.

MODEL IDEA A model **score** can rank **prediction**s without matching the real frequency of correct outcomes.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

### How the mechanism works

UNDER THE HOOD For this example, each **score** estimates the probability that the selected label is correct. Ten **prediction**s have **score**s near 0.8; six are correct, so observed correctness is 6/10=**60%**.

If instead **score**s mean probability of the spotted class, compare them with the fraction of reference-spotted leaves.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

### Step 1

STEP 1 Each **score** estimates the probability that the selected label is correct. Ten

**prediction**s have such **score**s near 0.8.

### Step 2

STEP 2 Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.

### Step 3

STEP 3 The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.

### Step 4

STEP 4 Use bins across many cases and design an uncertain zone requiring human review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a **score**.

Recalculate, retrace, or restate the **evidence** independently:

**Concrete teaching case**

### The teaching case

Ten cases receive probability .8 for spotted; eight are spotted.

### The teaching case

Interpret **calibration** and its limits.

### Exact explanation to model

> **Why the result follows**
>
> 8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

### Explain the added worked example

## Explain the added worked example · 1

If many comparable cases receive probability .8, a calibrated model should produce positives in roughly 80 percent of those cases over adequate data.

It does not mean every case is 80 percent correct or that exactly eight of each next ten must be positive. **Calibration** is assessed across groups.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **calibration** as a group property.

## Explain the added worked example · 2

In a fictional held-out check, ten cases all have predicted probability .2 and two are positive. Another ten all have probability .8 and six are positive.

Average **prediction**s are .2 and .8; observed frequencies are 2/10=.2 and 6/10=.6. The second group is overconfident for positives in this sample.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

### Key terms

| Term | Meaning |
| --- | --- |
| Calibration | Agreement between predicted probabilities and outcome frequencies across comparable groups |

| Term | Meaning |
| --- | --- |
| Bin | A declared interval grouping predictions |
| Observed frequency | Positive outcomes divided by cases in a group |
| Reliability diagram | A plot comparing group average prediction with observed frequency |

### Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

#### Calibration as a group property at a glance

| PART | EVIDENCE OR ACTION | INTERPRETATION |
| --- | --- | --- |
| Low probability group | Average .2 | Observed 2 of 10 |
| High probability group | Average .8 | Observed 6 of 10 |
| Compare groups | Prediction versus frequency | Keep group sizes visible |

Plot average prediction on the horizontal axis and observed frequency on the vertical axis. The two points are (.2,.2) and (.8,.6). Agreement lies along the diagonal. Use the average score within a bin, not automatically the bin midpoint. Small groups make frequencies uncertain.

**Which worked group overstates positive frequency and by how much? Explain the steps or evidence behind your response.**

| Thinking move | Teacher prompt |
|---|---|
| Plan | What is given? What is the question? Which procedure fits these facts? |
| Do | Point to each input and intermediate step in the worked trace. Name the rule that allows the next step. |
| Check | Does the output answer the question? Which assumption or limitation prevents a larger claim? |

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

### A pictorial worked case: Scores and Calibration Intuition

Treat **calibration** as a comparison across groups of probability **score**s.

### The facts we will use

#### Supplied facts

Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

#### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/14

# Scores and Calibration Intuition

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Group scores

Keep the ten 0.8 cases and ten 0.2 cases separate.



## 2. Compare frequencies

Observed spotted fractions are 8/10=0.8 and 4/10=0.4.



## 3. Interpret gap

The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

**CHANGE ONE THING**

The 0.2 group has two spotted cases instead of four.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Group scores | Keep the ten 0.8 cases and ten 0.2 cases separate. |
| 2. Compare frequencies | Observed spotted fractions are 8/10=0.8 and 4/10=0.4. |
| 3. Interpret gap | The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

#### Keep the stated limit

Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

### Change one thing in the pictured case

#### Change one thing

Change: The 0.2 group has two spotted cases instead of four.

#### Predicted result

**Prediction**: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

#### Why the result follows

Reason: **Calibration** compares stated probabilities with group outcome frequencies.

### Ask while showing the picture

## Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

# Demonstrate and guide practice

> Demonstrate Scores and Calibration Intuition, reduce support deliberately and check individual reasoning before discussing answers.

### Board sequence

#### 1. Put the facts on the board

Write the given facts: Ten cases receive probability .8 for spotted; eight are spotted.

#### 2. State the decision to explain

Write the question to explain: Interpret **calibration** and its limits.

#### 3. Reveal the worked reasoning

Reveal after **prediction**s: 8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

#### Hold back the independent task

Keep for an independent attempt: Among 20 **prediction**s with **score** 0.8, 12 are correct. What is observed accuracy in this group and how does it compare with 0.8?

### Classroom dialogue

#### Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

### Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

### Model these guided questions

Guided response: The .8 group exceeds observed .6 by .2, or 20 percentage points in this sample. Exit response: The reported improvement reuses the answers used for adjustment; evaluate the frozen adjustment on separate data.

### Case-specific teaching sequence

Stage · Teacher action

**Stage:** Session 1 start to 7 minutes

**Teacher action:** Use the prerequisite question: Does a score necessarily represent a probability? Check the explanation before proceeding.

**Stage:** 7 to 18 minutes

**Teacher action:** Read the opening and model the worked evidence. Ask students to identify calibration in context.

**Stage:** 18 to 32 minutes

**Teacher action:** Complete W 1 and W 2 with support: compute frequencies; locate the discrepancy. Require the evidence, not only the final response.

**Stage:** 32 to 45 minutes

**Teacher action:** Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

**Stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve the distinction between calibration and bin. Invite a counterexample to an overconfident claim.

**Stage:** 8 to 20 minutes

**Teacher action:** Discuss the application and complete W 3: Why can constant .5 predictions look calibrated but rank poorly?

**Stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently before giving hints. Assess these outcomes: Interpret calibration using groups of forecasts; Compare average predicted probability with observed frequency. Record any support used.

**Stage:** 35 to 45 minutes

**Teacher action:** Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

**Reduce help deliberately**

### Completion practice

Completion task: Which worked group overstates positive frequency and by how much?

### Reasoning to listen for

Teacher check: Teacher explanation: The .8 group exceeds observed .6 by .2, or 20

percentage points in this sample.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

### Individual reasoning check

Use the worked case for **Calibration** as a group property. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Expected repair

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: 2/10=.2 and 6/10=.6.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

### Pictorial practice question

Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Reasoned teacher answer

No. It describes a group proportion: here eight of ten are spotted, and two are not. It is not an individual guarantee. Changed-case check: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

**Calibration** compares stated probabilities with group outcome frequencies.

Changed-case reasoning: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. **Calibration** compares stated probabilities with group outcome frequencies.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

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TEACHER GUIDE · 4 OF 9

## Run the weekly evidence lab

> Run the Scores and Calibration Intuition investigation and use the first unsupported step to choose a teaching response.

Keep the test conditions fixed, record every result, and change only one planned factor.

1. Write the hypothesis and expected **evidence** before running.

2. **Reliability** Lab: sort thirty **prediction**s into **score** bins, calculate observed correctness, and identify overconfident regions.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

| Record | What to capture |
|---|---|
| Input or evidence | Exact data, instruction or claim |
| Expected result | Answer or safe fallback declared before the run |
| Observed result | Actual result, including errors |
| Explanation | Rule, calculation or evidence explaining the outcome |
| Next test | One justified change and a new prediction |

### Diagnose and reteach

MISCONCEPTION TO REPAIR A **score** of 0.9 means the **prediction** has exactly a 90% chance of being correct.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

### Address these additional misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: 2/10=.2 and 6/10=.6.

A stronger claim makes a stronger project: The reported improvement reuses the answers used for adjustment; evaluate the frozen adjustment on separate data.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Points (.3,.4) and (.7,.7). The first understates **observed frequency** by .1; the second agrees on this sample.

Group sizes still limit claims.

### Match the support to the first error

| What the response shows | Next teaching action |
|---|---|
| Missing or misread facts | Read the case aloud, mark supplied versus unknown facts, and let the learner restate it. |
| Wrong procedure | Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation. |
| Arithmetic or implementation error | Trace intermediate values or lines; preserve the first mismatch and retry that step. |
| Correct result without a reason | Ask for a trace and a changed case before marking independent understanding. |
| An unsupported wider claim | Ask which supplied evidence supports the claim and which further observation would be needed. |

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

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## Assess and explain the answers

> Use the Scores and Calibration Intuition questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

### Worked case answer

#### Teacher key · Worked case answer

8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

### Independent question

#### Independent question · withhold the key

Among 20 **prediction**s with **score** 0.8, 12 are correct. What is observed accuracy in this group and how does it compare with 0.8?

### Expected independent answer

#### Teacher key · Expected independent answer

Observed accuracy is 12/20 = **60%**, below the **80% score**. This small group suggests overconfidence, but more data is needed to assess **calibration** reliably.

### What counts as evidence

| Criterion | 0 points | 1 point | 2 points |
|---|---|---|---|
| Result | Incorrect or absent | Partly correct | Correct with necessary conditions |
| Reasoning | No supporting trace | Some steps | Clear mechanism, calculation or source support |
| Limit | Overclaims certainty | Vague limitation | Relevant uncertainty or missing fact |
| Transfer | Copies without adapting | Adapts with prompts | Solves a changed case independently |

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

**Feedback**

Ask the learner to find the **evidence** that challenges this claim: “A model **score** of 0.8 guarantees that this particular **prediction** is correct.” Keep the first attempt and an uncoached retry separate.

**Original transfer answer**

### Teacher key · Original transfer answer

14/20=.7 matches the stated **score** for this small group, illustrating **calibration** if **score**s mean estimated correctness probability. It does not establish global **calibration**, stability or certainty for one case.

Examine more data and other bins with a clear **score** meaning.

**Explanations for the additional workbook examples**

**W1 Compute frequencies**

### Workbook question · W1

Give the observed frequencies for the two worked groups.

### Teacher reasoning · W1

Teacher explanation: 2/10=.2 and 6/10=.6.

### Teacher reasoning · W1

Use each group’s observed positive count over its observed total: 2/10 = 0.2 and 6/10 = 0.6. These are sample frequencies for the worked groups.

Do not turn either small-group **observation** into a guarantee about the next individual or a proof of **calibration**.

### W2 Locate the discrepancy

#### Workbook question · W2

Which worked group overstates positive frequency and by how much?

#### Teacher reasoning · W2

Teacher explanation: The .8 group exceeds observed .6 by .2, or 20 percentage points in this sample.

### W3 Separate ranking

#### Workbook question · W3

Why can constant .5 **prediction**s look calibrated but rank poorly?

#### Teacher reasoning · W3

Teacher explanation: When half the cases are positive, the overall fraction agrees, but every case has the same **score** and no ranking is provided.

### W4 Apply independently

#### Workbook question · W4

Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

### Teacher reasoning · W4

Teacher explanation: Points (.3,.4) and (.7,.7). The first understates **observed frequency** by .1; the second agrees on this sample. Group sizes still limit claims.

**W5 Protect evaluation**

### Workbook question · W5

What is wrong with calibrating and reporting success on the same final-test records?

### Teacher reasoning · W5

Teacher explanation: The reported improvement reuses the answers used for adjustment; evaluate the frozen adjustment on separate data.

**Assess independent transfer**

### Independent question · withhold the key

Selected case: Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

**Selected case answer**

### Teacher key · Selected case answer

Teacher explanation: Points (.3,.4) and (.7,.7). The first understates **observed frequency** by .1; the second agrees on this sample. Group sizes still limit claims.

| Dimension | Evidence required |
| --- | --- |
| Explanation | Names the applicable idea or procedure and ties it to the supplied facts. |
| Trace and result | Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution. |
| Limits | States the assumptions, missing evidence or conditions under which the result would change. |

### Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

### Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

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## Connect and extend learning

> Connect Scores and Calibration Intuition to Responsible Plant Classifier, adapt support and plan a later changed-case check.

### Responsible Plant Classifier

Add an uncertainty band to the Plant Classifier and define what the interface says when **score**s fall inside it.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

### Support and stretch

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

### Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

### Bridge to the next week

Next: Week 15: Confusion Matrix. Retrieve today's **evidence** and identify what carries forward and what is new.

### Support and fresh reassessment

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **calibration** and bin in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Points (.3,.4) and (.7,.7). The first understates **observed frequency** by .1; the second agrees on this sample. Group sizes still limit claims.

Fresh retry: five .6 **prediction**s with two positives give (.6,.4), an overstatement of .2 in a very small group. Require a small-sample limitation. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Next week: Error analysis and controlled follow-up. Carry forward the saved **evidence** and the distinction between development choices and justified claims.

### Plan the delayed return

Suggested return points in this level: ai-4/15, ai-4/17, ai-4/20. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

| Record | Teacher evidence |
|---|---|
| Learner code and date | |
| Concept and case used; prior exposure | |
| Explanation / trace / limits | |
| Support and first repair | |
| Changed case and delayed outcome | |
| Next teaching action | |

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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## Prepare, teach and assess

> Use this preparation summary alongside the detailed Scores and Calibration Intuition lesson; it is a checklist rather than a replacement for the full sequence.

### Teaching target

Teaching target: Interpret **calibration** using groups of forecasts

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

**Session:** Session 1 · Understand · 45 min

**Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

**Session:** Session 2 · Practise · 45 min

**Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

**Session:** Session 3 · Investigate and transfer · 45 min

**Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Which worked group overstates positive frequency and by how much?

### Supported practice answer

Reasoned practice answer: Teacher explanation: The .8 group exceeds observed .6 by .2, or 20 percentage points in this sample.

Check every learner before discussion: Use the worked case for **Calibration** as a group property. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

### Repair the misconception

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: 2/10=.2 and 6/10=.6.

### Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

### Extend the reasoning

Extension: The 0.2 group has two spotted cases instead of four. Require a **prediction** and a reason before comparing. Expected result: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

Reason: **Calibration** compares stated probabilities with group outcome frequencies.

### Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

| Criterion | Not yet | With support | Independent |
|---|---|---|---|
| Explanation | Names an answer without a deciding fact | Uses a hint to link fact and decision | Explains the deciding fact in own words |
| Trace or test | Cannot connect input and result | Completes steps with prompts | Traces or tests a changed case accurately |
| Limits and revision | Claims more than evidence supports | Names a limit after a prompt | States a defensible limit and useful next test |

### Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

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## Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery **score**s. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

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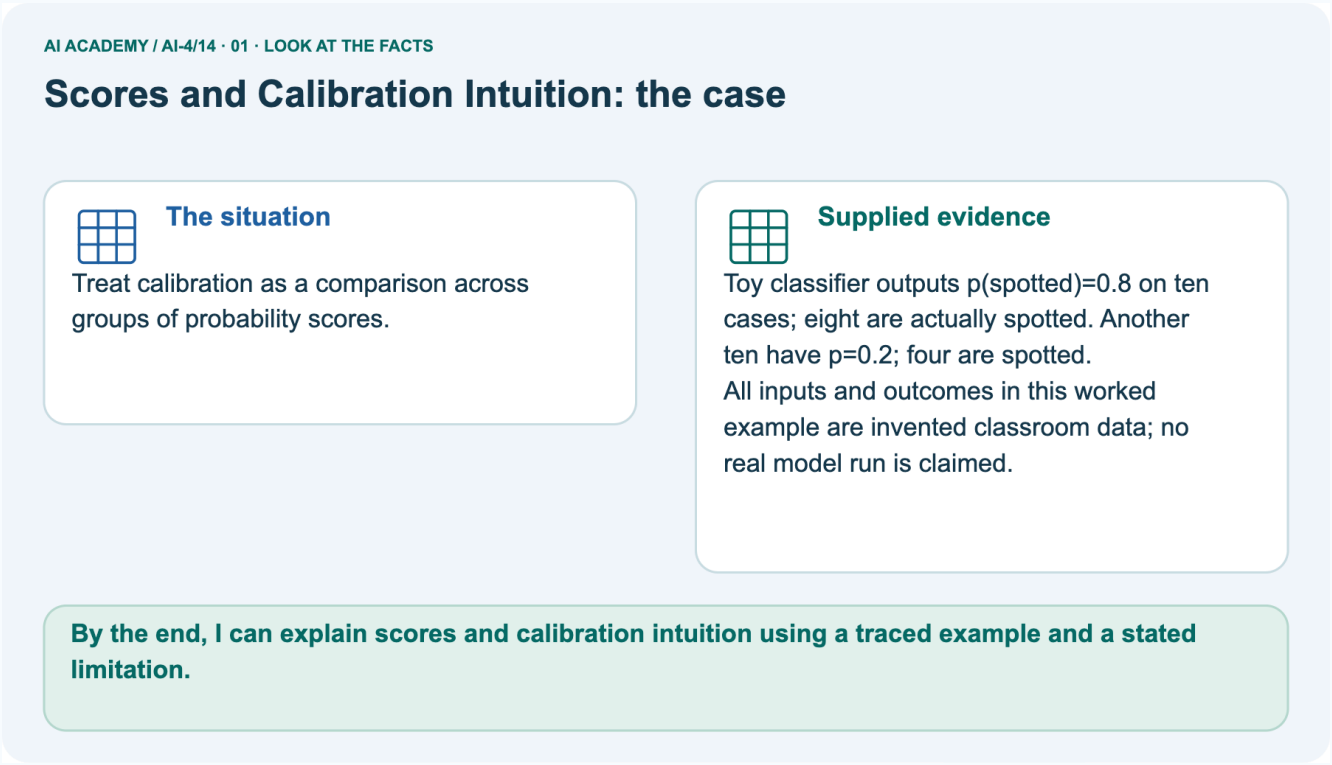


## Visual concept journey

> Show one Scores and Calibration Intuition graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

## 01. Scores and Calibration Intuition: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

Treat **calibration** as a comparison across groups of probability **score**s. Supplied facts: Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

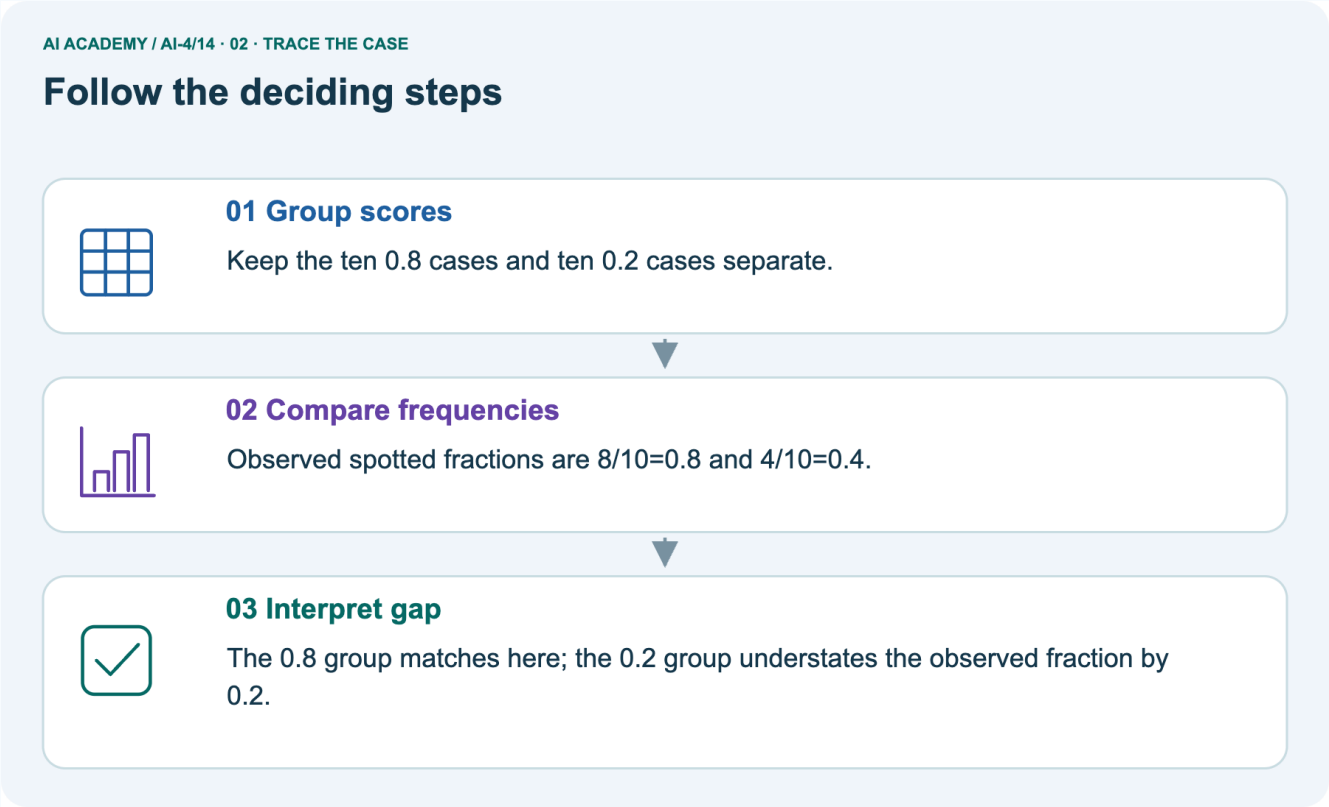
Point to the given fact that matters most. Explain why it matters.

### Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the

image is difficult to see.

### 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

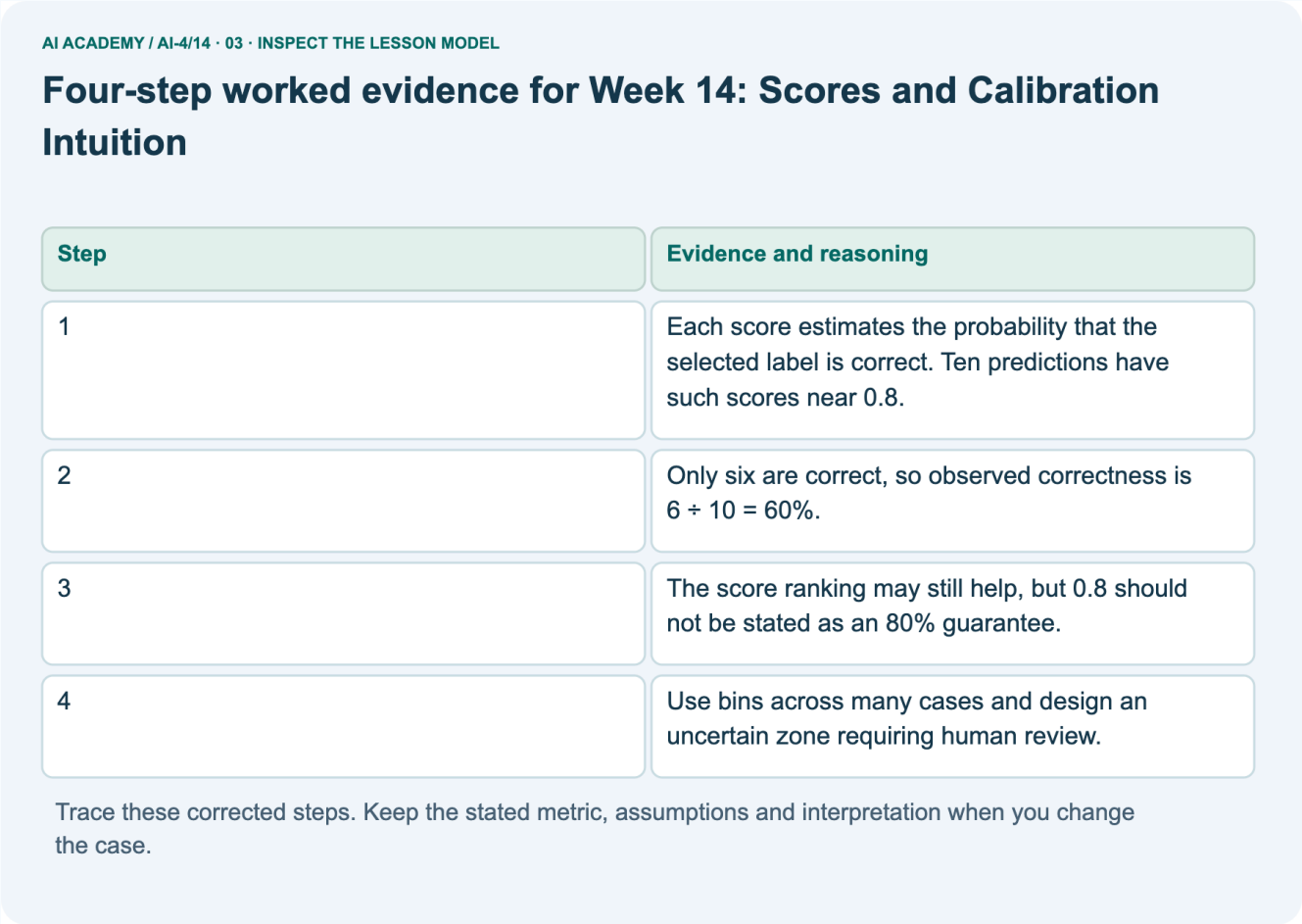
Group **score**s: Keep the ten 0.8 cases and ten 0.2 cases separate. → Compare frequencies: Observed spotted fractions are 8/10=0.8 and 4/10=0.4. → Interpret gap: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

Point to each numbered stage and explain which recorded **input** it uses.

### Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

### 03. Four-step worked evidence for Week 14: Scores and Calibration Intuition




Follow the labelled picture with the matching explanation. Open the vector diagram

Four-step worked **evidence** for Week 14: **Score**s and **Calibration** Intuition. 1 | Each **score** estimates the probability that the selected label is correct.

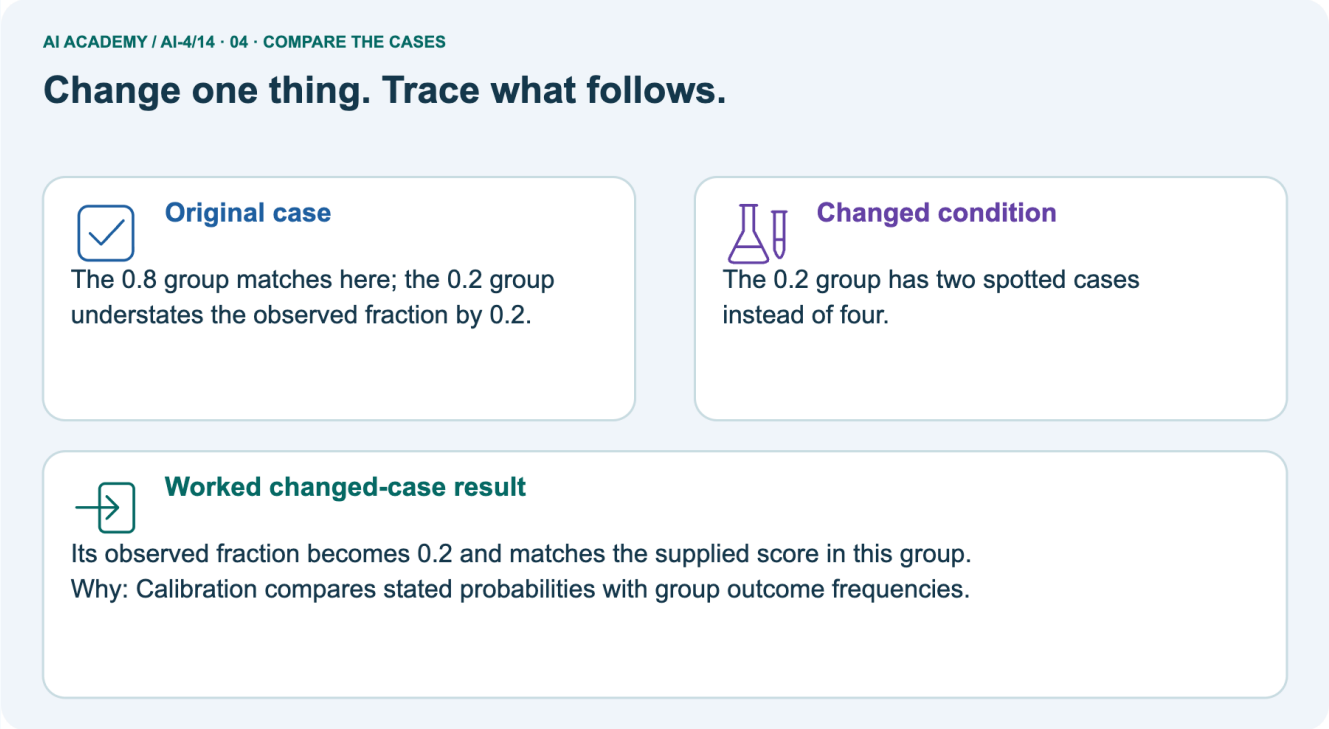
Ten **prediction**s have such **score**s near 0.8.. 2 | Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.. 3 | The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.. 4 | Use bins across many cases and design an uncertain zone requiring human review.

Read the labels and values. Explain the relationship the model shows.

### Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

**04. Change one thing. Trace what follows.**




Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.
Changed condition: The 0.2 group has two spotted cases instead of four.

Expected result: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.
Reason: **Calibration** compares stated probabilities with group outcome frequencies.

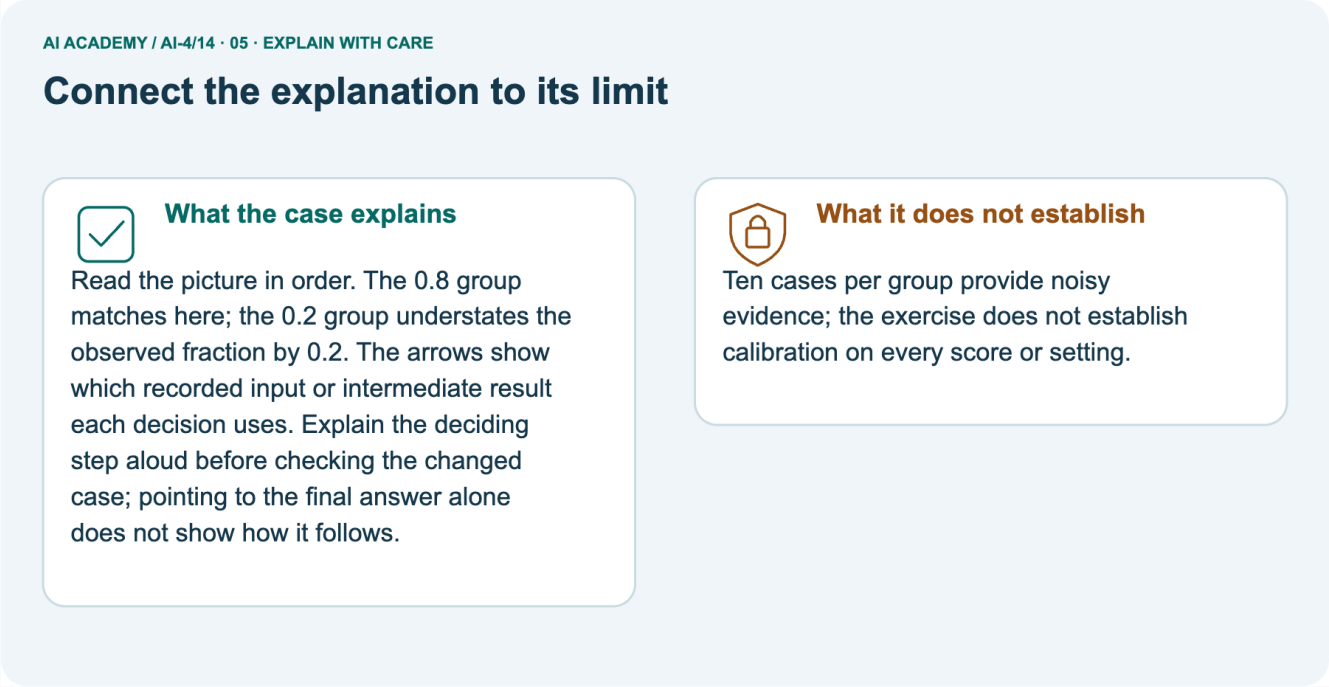
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Calibration** compares stated probabilities with group outcome frequencies.

### 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

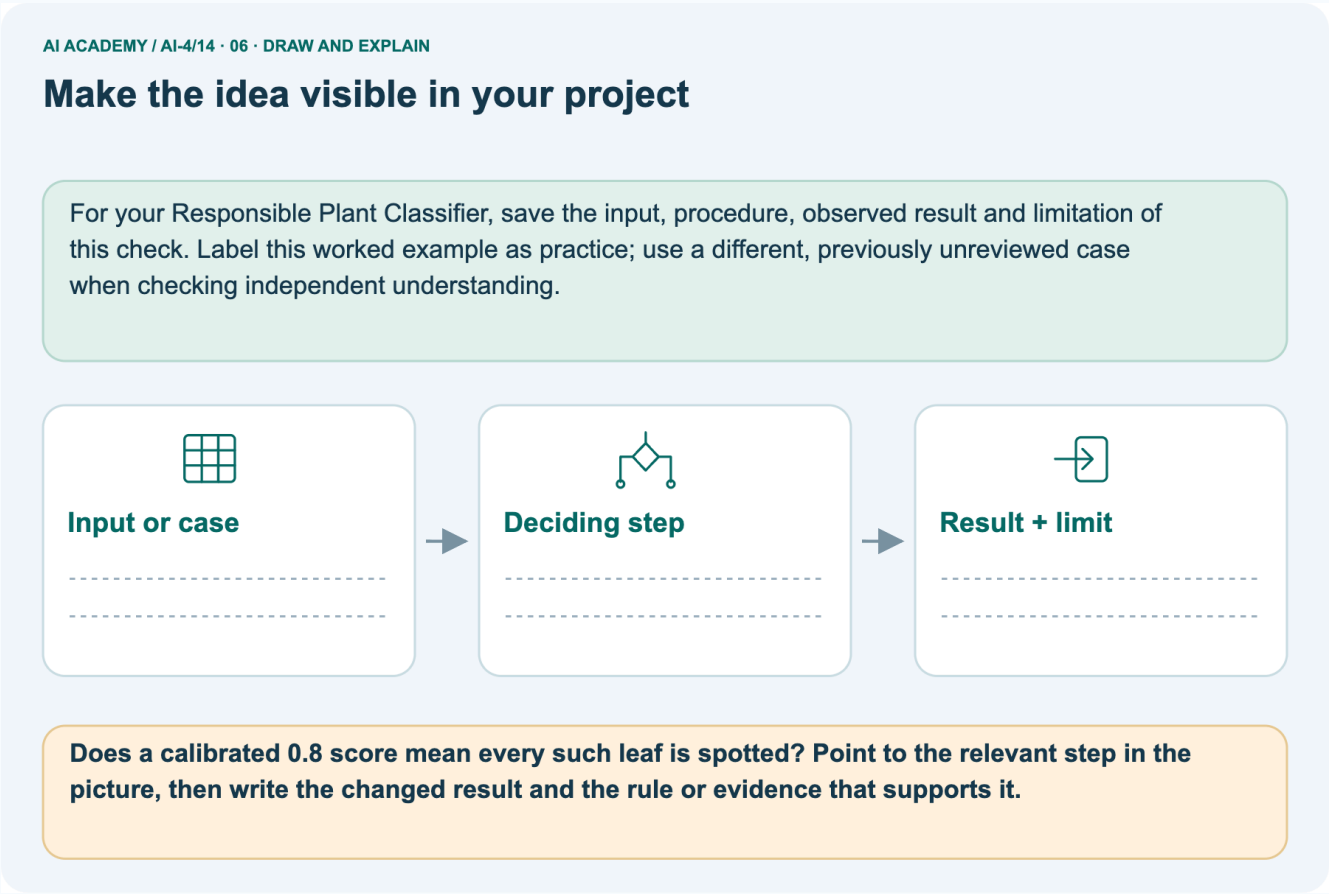
Explain the case, then point to a limit on the claim.

### Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

### Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. It describes a group proportion: here eight of ten are spotted, and two are not. It is not an individual guarantee.

Changed-case check: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. **Calibration** compares stated probabilities with group outcome frequencies.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.